

Education

- 2025–pres. Harvard University, Cambridge, MA, PhD in Physics
- 2020–2024.5 Brown University, Providence, RI. B.Sc. in Mathematical Physics
Thesis: Reconstructing Spin Liquids
- 2014–2020 The Overlake School, Redmond, WA

Publications

- 2026 **L. Z. Brito**, C. Zhang, A. Vishwanath. *Wen–Zee shift in the stack and condense construction of fractional quantum hall hierarchy*, preprint. [forthcoming].
- L. Z. Brito**, S. D. Pace. *Lattice 2-group symmetries: operators, defects, and gauging*, preprint. [[arXiv:2609.10682](https://arxiv.org/abs/2609.10682)].
- L. Z. Brito**, R. P. Cabeen, D. H. Laidlaw. *General microstructure factor analysis of diffusion MRI in gray-matter predicts cognitive scores*, NeuroImage. [[neuroimage.2026.121991](https://doi.org/10.1016/j.neuroimage.2026.121991)].
- 2025 **L. Z. Brito**, J. B. Marston. *Reconstructing Spin Hamiltonians of 2D Gutzwiller-Projected Wavefunctions*, preprint. [[arXiv:2510.15034](https://arxiv.org/abs/2510.15034)].
- 2024 **L. Z. Brito**, S. Carr, J. A. Jacoby, J. B. Marston. *Hamiltonian Reconstruction: the Correlation Matrix and Incomplete Operator Bases*, Physical Review B. [[PhysRevB.110.125118](https://doi.org/10.1103/PhysRevB.110.125118)].
- 2023 D. Candoli, I. K. Nikolov, **L. Z. Brito**, S. Carr, S. Sanna, V. F. Mitrović, *PULSEE: A software for the quantum simulation of an extensive set of magnetic resonance observables*, Computer Physics Communications. [[doi:j.cpc.2022.105898](https://doi.org/10.1016/j.cpc.2022.105898)].

Employment

- 2023–2025 *Instructional Lab Assistant, Brown Department of Physics*
Specialization in quantum optics. Designed and implemented tabletop demonstration of Bell's inequality, single photon interferometry. Wrote Python photon acquisition software for laboratory usage.
- 2022–2023 *Teaching Assistant, APMA1930W - Probabilities in Quantum Mechanics*
Wrote and lectured supplementary curriculum on quantum information theory and related topics. Substitute lectured to 30+ students.
- 2017–2018 *Assistant instructor, Play-Well Teknologies*

Awards and Recognition

- 2025–2026 *Harvard Purcell Fellowship*.
- 2025 *R. Bruce Lindsay Prize for Excellence in Physics*. Awarded to one Brown Undergraduate each year for significant contributions to the physics department.
- 2023–2024 *Goldwater Scholar in Physics and Astronomy*. National scholarship awarded to 300 of the nation's top STEM undergraduates.
- 2023 *Karen T. Romer Undergraduate Teaching and Research Award*. Competitive grant awarded to Brown undergraduates to conduct summer research.

Relevant Coursework

Physics

- Advanced Electromagnetic Theory
- Quantum Mechanics I*, II*
- Advanced Quantum Mechanics*
- Advanced Statistical Mechanics*
- Computational Physics*
- Quantum Theory of Fields I*, II*
- Quantum Many-Body Theory*
- Gauge Theory*
- Standard Model*
- Biophysics*

Computer Science

- Accelerated Intro to Computer Science
- Data Science
- Intro to Software Engineering
- IS in Biophysical Scientific Visualization
- Interdisciplinary Scientific Visualization*
- Deep Learning*

Mathematics

- Linear algebra
 - Partial Differential Equations
 - Abstract Algebra
 - Complex Analysis
 - Differential Geometry*
- (* = graduate level)